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AttriSelect: Efficient Attribute-Driven Data Selection for Targeted Vision Model Repair

(/pdf?id=JxkGmySuLb)

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TL;DR: Attribute-Driven Data-Centric Bug Discovery and Repair

Abstract:
Targeted vision repair aims to correct model failures on rare semantic slices after a debugging pipeline identifies the failure slice and a generator produces corresponding repair images. In most existing workflows, these repair images are appended to the original training set, and the model is retrained or fine-tuned using all source data. While this preserves broad coverage, it also makes each repair cycle computationally expensive and time-consuming, especially as repairs accumulate over multiple iterations. Moreover, it provides little control over which source examples actually contribute to the repair. This poses a data selection problem related to active learning: rather than using the full dataset, the goal is to identify a compact yet informative subset that best supports repair while substantially reducing retraining cost. However, an effective repair subset must satisfy requirements beyond standard sample efficiency. In particular, it should preserve class coverage, retain rare repair-relevant attributes, and avoid redundancy with the generated repair data. Existing active learning methods typically prioritize uncertainty, diversity, or gradient-space coverage, but these signals are often poorly aligned with the semantic attributes that define the failure slice. To address this limitation, we propose AttriSelect, an attribute-driven selector that prioritizes samples based on attribute rarity, generated-data coverage, and soft class balance. We provide theoretical analysis and evaluate AttriSelect across image classification, pose estimation, and object detection. Across tasks and repair budgets, AttriSelect gives the strongest repair results in most settings, showing that explicit attribute metadata provides a repair-specific selection signal distinct from traditional active learning criteria.

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Meta Review of Submission12417 by Area Chair 6k27

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Metareview:

Summary: This paper proposes to use attributes that the existing dataset fails to cover to select repair data from the source data pool or generated data. The authors evaluate the method across tasks, including image classification, pose estimation, and object detection. Results suggest that the proposed method yields better repair effects than the compared baselines.

Strengths: The paper provides a unique perspective on selecting data for model repair. Attribute-based selection is reasonable and yields a strong performance compared with other approaches. Since the upstream generation process produces attributes, it is also reasonable to leverage these attributes.

Weaknesses: I list two major weaknesses that significantly influence my assessment of this submission:

- Reviewers yyPZ and hG6j raise the concern of the end-to-end cost and misleading claims about efficiency. It is also related to another shared concern about fair comparison against baselines. As stated in the above strengths section, it is very reasonable to leverage the labeled attributes since the upstream provides them. However, it seems that none of the compared baselines use these attributes.

  Given attributes, the proposed method is actually not restricted to model repair, but it can be used in any data selection scenario. The difference is that the upstream has provided attribute labels in this paper, while for standard data selection, attribute generation will usually be counted into the method's computation. The authors ingeniously pick this setting so that it cannot be strictly blamed for discarding attribute generation when calculating computational overhead.

  While the reported efficiency cannot be directly criticized, I do agree that the adopted baselines should include methods considering the generated attribute labels. The authors should accordingly include attribute-aware baselines as suggested by the reviewers. Otherwise, this is merely a standard data selection method that should be evaluated outside of the model repair scenario. It would further strengthen the paper if the authors indeed include such comparisons on standard data selection scenarios.

  Another possibility is that the compared baselines should also use the attribute values instead of extracting embeddings. This will make the efficiency comparison fairer.
- Reviewers hG6j and KaLa raise the question regarding different repair pipelines. Since AttriSelect itself only involves selecting from the source pool and generated data, it would be stronger if the authors could evaluate the model against multiple pipelines. The authors should clarify the implementation details regarding the entire repair pipeline. When the ablation on generative models has been provided, it would further strengthen the paper if the authors could also ablate the attribute generation and other pipeline components. The robustness against imperfect attribute vocabularies will also be important for the pipeline.

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Response to Area Chair 6k27 (Part 1)

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by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:16 (modified: 30 Jul 2026, 02:51)

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Comment:

Dear Area Chair 6k27, thank you for consolidating the reviewers' concerns.

This response is organized into 5 parts. We focus on evaluation scope and the limits of our claims. Our responses to Reviewers yyPZ, hG6j, and KaLa contain the full protocols; here we collect the relevant evidence and conclusions.

1. End-to-End Cost

Table 2 of the submitted paper measures source-subset selection after attribute records are available; the reported `39––283x` speedups apply only to that stage. The accounting below separates source assignment, selection, retraining, and evaluation. We report repair-image generation separately because every selector uses the same fixed, publicly available generated set.

Dataset	Source Images	Assigned Attributes	H200 Assignment Time	Normalized Assignment Time (min / 1K images / attribute)
KITTI	5,000	6	32.77 min	1.09
ImageNet10	10,400	3	29.35 min	0.94
COCO	20,000	5	90 min	0.90

`Normalized Assignment Time` divides the total H200 assignment time by the source-pool size in thousands and the number of assigned attributes. For example, the COCO value is `90 / (20 x 5) = 0.90` minutes per 1,000 images per attribute.

The first-run totals below conservatively charge fresh full-pool VLM assignment in every AttriSelect row.

KITTI end-to-end wall-clock time.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect
0.1	42.17 min	42.18 min	48.82 min	1.37 h	1.12 h	43.47 min	43.53 min	1.01 h
0.3	43.84 min	43.85 min	50.28 min	1.44 h	1.20 h	45.08 min	45.63 min	1.09 h
0.5	45.18 min	45.18 min	51.73 min	1.49 h	1.27 h	46.42 min	47.47 min	1.11 h
0.7	47.04 min	47.05 min	53.46 min	1.56 h	1.35 h	48.16 min	49.73 min	1.13 h
0.9	48.86 min	48.87 min	55.53 min	1.63 h	1.42 h	50.34 min	51.46 min	1.15 h

COCO end-to-end wall-clock time.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect
0.1	4.41 min	4.42 min	10.98 min	16.18 min	19.63 min	5.08 min	6.97 min	1.57 h
0.3	8.18 min	8.18 min	14.74 min	39.86 min	49.76 min	8.93 min	17.86 min	1.62 h
0.5	12.37 min	12.36 min	19.04 min	1.08 h	1.56 h	13.08 min	29.47 min	1.66 h

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect
0.7	15.73 min	15.75 min	22.27 min	1.45 h	2.29 h	16.34 min	37.63 min	1.69 h
0.9	19.66 min	19.65 min	26.28 min	1.87 h	3.01 h	20.18 min	40.94 min	1.72 h

Cost takeaway. The faster method depends on the source ratio and whether fresh attribute assignment is charged. **Random, Stratified, UHerding, and PPAL cost less, especially at small ratios, but usually repair less well.** At COCO ratio 0.1, for example, AttriSelect takes 1.57 hours with fresh annotation included. Random, UHerding, and PPAL take 4.41, 5.08, and 6.97 minutes, but reach only 0.6048, 0.6013, and 0.5950 AP, versus 0.6183 for AttriSelect.
CoreSet and BADGE are usually the strongest non-AttriSelect baselines, but their costs rise sharply with the source ratio. At ratio 0.9, both are slower than AttriSelect even though the AttriSelect total includes VLM annotation of the full source pool. On COCO, AttriSelect takes 1.72 hours and reaches 0.6355 AP, compared with 1.87 hours/0.6202 for CoreSet and 3.01 hours/0.6213 for BADGE. On KITTI, the corresponding values are 1.15 hours/0.5850 mAP50--95 for AttriSelect, 1.63 hours/0.5781 for CoreSet, and 1.42 hours/0.5810 for BADGE. `At ratio 0.9 on both COCO and KITTI, AttriSelect is more accurate and faster on the first repair run, even after charging full-pool VLM assignment.`

These totals are conservative because each AttriSelect row pays the full source-assignment cost. `In later repair cycles, the same metadata are reused, and the online AttriSelect computation takes less than two seconds.` We will relabel Table 2 as source-subset selector runtime, add this cost decomposition, and state explicitly that `the reported 39--283x speedups begin after the attribute data are available` .

2. Same-Information Repair Baselines

We compare AttriSelect with three controls that receive the same ImageNet10 attributes and the same `pink AND fabric` target definition: Target-Slice + Stratified, Attribute-Filtered Random, and Attribute-Filtered Entropy. All methods use the same fixed 1,000 generated images, source budget, checkpoint, 15-epoch schedule, and five seeds.

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Response to Area Chair 6k27 (Part 2)

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Official Comment by Authors 29 Jul 2026, 22:16 (modified: 30 Jul 2026, 02:51)
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Comment:
Five-seed ImageNet10 same-information repair comparison. Metric: test accuracy (fraction; mean ± standard deviation).

Source Ratio	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.1	0.7303 ± 0.0051	0.7340 ± 0.0034	0.7317 ± 0.0025	**0.7369 ± 0.0043**
0.3	0.7385 ± 0.0024	0.7382 ± 0.0042	0.7365 ± 0.0023	**0.7492 ± 0.0044**
0.5	0.7438 ± 0.0037	0.7442 ± 0.0038	0.7442 ± 0.0031	**0.7508 ± 0.0010**

Source Ratio	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.7	0.7434 ± 0.0065	0.7474 ± 0.0038	0.7482 ± 0.0020	**0.7515 ± 0.0037**
0.9	0.7485 ± 0.0026	0.7469 ± 0.0036	0.7503 ± 0.0045	**0.7562 ± 0.0019**

Key result. `With attribute metadata and the target-slice definition matched across methods, AttriSelect has the highest mean accuracy at all five source ratios.` Its margin over the strongest control ranges from 0.29 to 1.07 accuracy points and averages 0.59 points. The attributes alone do not account for the difference.

3. Standard Source-Only Data Selection

We evaluate source-only ImageNet10 data selection outside model repair. There are no generated images, diagnosed slices, or repair checkpoints. Each selected subset trains a fresh ResNet-18, and both tables report five-seed mean test accuracy.

The comparison adds three attribute-aware baselines absent from the submitted paper: Attribute Rarity, Semantic CoreSet, and Object--Attribute Entropy Reduction [2]. **Attribute Rarity.** We score each image by the class-conditional rarity of its VLM-assigned attribute values and apply soft class balancing. In Active Learning, rarity counts use the labeled set and candidate classes use model predictions.

Semantic CoreSet. We one-hot encode the VLM-assigned attribute values, assign an exact class-balanced quota, and run greedy k-center selection within each class in this semantic representation. The implementation selects without replacement.

Object--Attribute Entropy Reduction [2]. Since the attributes are already known, we adapt the method to query only the unknown class label. It prioritizes samples with high model class entropy and under-labeled attribute values whose labeled class distributions remain uncertain, followed by soft class balancing.

One-Shot Data Selection

Five-seed ImageNet10 source-only one-shot data selection. Metric: test accuracy (%).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	Attribute Rarity	Semantic CoreSet	AttriSelect
0.1	43.23	41.85	26.86	38.33	42.62	44.46	44.08	**45.00**
0.3	55.84	57.15	44.64	56.16	58.17	56.54	56.55	**58.32**
0.5	64.77	64.14	58.69	64.87	63.19	65.31	62.56	**66.53**
0.7	69.47	71.63	66.71	68.03	69.57	71.01	67.18	**72.02**
0.9	71.34	71.70	72.88	72.89	72.38	72.48	71.24	**72.92**

Takeaway. `AttriSelect has the highest accuracy at all five one-shot budgets.` At source ratios `0.1/0.3/0.5/0.7/0.9` , it exceeds the strongest competing method by `0.54/0.15/1.22/0.39/0.03` accuracy points, respectively. The advantage is largest at ratio `0.5` and remains positive even when `90%` of the source pool is retained.

3. Standard Source-Only Data Selection (continued)

Iterative Active Learning

Five-seed ImageNet10 source-only iterative Active Learning. Metric: test accuracy (%); `Curve Mean` averages the six labeled ratios `0.1`–`0.6`.

Ratio	Random	Entropy	CoreSet	BADGE	Attribute Rarity	Object--Attribute Entropy Reduction [2]	AttriSelect-AL
0.1	43.23	43.23	43.23	43.23	43.23	43.23	43.23
0.2	51.11	**52.26**	50.00	51.27	50.93	50.48	51.33
0.3	55.38	54.85	55.01	56.17	52.94	55.95	**56.96**
0.4	58.71	58.63	59.54	60.46	60.47	60.31	**63.02**
0.5	61.97	63.70	63.92	61.89	61.86	63.88	**65.18**
0.6	66.15	66.70	68.16	67.11	65.78	65.79	**68.80**
Curve Mean	56.09	56.56	56.64	56.69	55.87	56.61	**58.09**

> **Takeaway.** All Active Learning methods share the same initial labeled set at ratio `0.1`. `AttriSelect-AL is strongest at ratios 0.3–0.6 and improves the Curve Mean by 1.40 points over BADGE.` This source-only experiment uses matched training and information, so the result does not depend on a model repair checkpoint.

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Response to Area Chair 6k27 (Part 3)

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by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

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Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs

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4. Alternative Repair Pipelines, Generators, and Implementation

`All submitted experiments place AttriSelect inside the SafeFix repair pipeline.` AttriSelect ranks only the original source pool, never the generated images or a joint pool. Every selector trains on its selected source subset plus the same fixed 1,000 generated images, with the checkpoint, optimizer, learning rate, epoch count, and source budget held fixed. `Only the source-selection rule changes.` The generator studies replace ControlNet with InstructPix2Pix on CelebA and ICEdit on COCO while leaving the rest of the pipeline unchanged.

Selection Interface and Repair Data

Selection pool. Let r denote the reported source-selection ratio. AttriSelect returns $\mathcal{D}^{sel} \subseteq \mathcal{D}^{src}$ with $|\mathcal{D}^{sel}| = \lfloor r|\mathcal{D}^{src}| \rfloor$. The generated set affects the score only through its attribute counts; generated images are not selection candidates.

`Changing the source ratio changes only the number and composition of real source images; the generated set remains fixed.`

> **Repair-pipeline summary.**
> **Upstream SafeFix:** diagnosis checkpoint $\theta_t \rightarrow$ diagnosed failure slice $T_{\text{bug}} \rightarrow$ controlled generation with ControlNet/Stable Diffusion 1.5, soft HED conditioning, and 30 DDIM steps $\rightarrow$ fixed $\mathcal{D}^{gen}$ of 1,000 images.
> **AttriSelect:** original $\mathcal{D}^{src}$ with attribute metadata plus attribute counts from $\mathcal{D}^{gen} \rightarrow$ source-only ranking $\rightarrow \mathcal{D}^{sel}$.
> **Repair training:** $\mathcal{D}^{sel} \cup \mathcal{D}^{gen} \rightarrow$ matched training or fine-tuning of θ_t at learning rate $1 \times 10^{-5} \rightarrow \theta_{t+1}$. ImageNet10 uses 15 repair epochs; CelebA, COCO, and KITTI use half of their corresponding diagnosis-stage schedules.

Dataset-specific implementation.

For ImageNet10, we train the diagnosis-stage ResNet-18 for 30 epochs using cross-entropy loss, Adam, learning rate 1×10^{-4}, weight decay 1×10^{-3}, and batch size 64. Qwen3-VL supplies closed-set attribute assignments for the source images. Repair then runs for 15 epochs at learning rate 1×10^{-5}. For CelebA, COCO, and KITTI, repair uses half as many epochs as diagnosis and the same repair learning rate, 1×10^{-5}.

HiBug Repair Pipeline

HiBug [3] is an attribute-based debugging and repair framework. It uses vision--language models to find interpretable failure slices and generate targeted repair examples. We test AttriSelect inside HiBug on CelebA. All selectors share the diagnosis checkpoint, 80,000-image source pool, fixed HiBug-generated repair set, source ratios, retraining configuration, and five seeds.

CelebA with the HiBug repair pipeline. Metric: five-seed mean FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect (HiBug)
0.1	4.59	2.33	6.92	-6.20	2.87	-1.36	5.58	**7.36**
0.3	5.91	5.08	8.02	7.79	1.66	3.39	3.87	**8.07**
0.5	5.76	7.70	7.48	7.90	2.53	2.13	9.64	**12.54**
0.7	2.57	8.06	10.03	6.19	7.86	4.96	5.90	**11.16**
0.9	2.84	4.28	7.95	5.71	5.80	4.19	6.37	**13.42**

> **Key result.** `Under HiBug, AttriSelect is strongest at all five source ratios and exceeds the strongest baseline by 2.00 FR points on average.` This experiment changes the repair pipeline. The InstructPix2Pix and ICEdit experiments below change only the generator within the SafeFix workflow.

Alternative Generators

CelebA with InstructPix2Pix, FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect (InstructPix2Pix)	AttriSelect (ControlNet)
0.1	4.23	1.95	6.41	-6.65	2.34	-1.78	5.12	**7.65**	8.48
0.3	5.34	4.72	7.98	7.21	1.02	2.81	3.45	**8.12**	8.91
0.5	5.56	7.23	6.78	7.54	2.12	1.56	9.12	**11.17**	13.79
0.7	2.12	7.67	9.34	5.78	7.34	4.23	5.34	**12.13**	12.09
0.9	2.45	3.92	7.56	5.34	5.23	3.87	5.91	**13.87**	14.74

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Response to Area Chair 6k27 (Part 4)

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COCO with ICEdit, FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect (ICEdit)	AttriSelect (ControlNet)
0.1	3.78	5.23	5.87	-0.34	2.34	3.12	1.56	**7.12**	6.86
0.3	3.92	8.67	8.12	9.12	6.56	4.67	6.89	**10.34**	9.86
0.5	7.91	8.45	9.67	6.91	7.82	6.54	7.23	**10.67**	10.13
0.7	6.54	10.12	9.98	9.56	9.12	6.71	8.23	**10.87**	10.18
0.9	5.12	9.87	6.98	7.82	8.12	7.54	10.12	**11.56**	11.05

> **Key result.** `AttriSelect is the strongest source selector at every evaluated CelebA ratio under InstructPix2Pix, exceeding the strongest baseline by 2.51 FR points on average.`

> **Key result.** Under ICEdit, AttriSelect is strongest at every evaluated COCO ratio and exceeds the best baseline by **1.13 FR points on average**. `Across the two generator substitutions, AttriSelect ranks first in all 10 evaluated dataset–ratio settings.`

Repair-Schedule Sensitivity

Matched protocol. We vary ImageNet10 repair training from 15 to 60 epochs while holding the diagnosis checkpoint, generated repair set, source budget, and five seeds fixed. The table compares AttriSelect with the strongest baseline in each ratio--epoch configuration.

Source Ratio	Method	15 Epochs	30 Epochs	45 Epochs	60 Epochs
0.5	Strongest baseline	10.47	10.74	11.84	12.11
0.5	**AttriSelect**	**11.85**	**12.39**	**13.51**	**14.08**
0.9	Strongest baseline	11.85	12.33	13.44	13.76
0.9	**AttriSelect**	**13.76**	**14.25**	**14.99**	**15.22**

Metric: five-seed mean FR (%). `AttriSelect is strongest in all eight ratio–-epoch configurations, so its advantage is not explained by the submitted paper's 15–epoch repair schedule.`

Scope of the Evidence

Scope. The empirical evidence covers one matched alternative repair pipeline (HiBug on CelebA) and two generator substitutions within the SafeFix-style workflow (InstructPix2Pix on CelebA and ICEdit on COCO).

The appendix theory gives a conditional argument for transfer across repair pipelines and retraining strategies. The result concerns source selection, not a particular pipeline or optimizer. Under the rarity-dominates-penalty condition, target-slice source examples remain ahead of non-target examples after accounting for the generated-data distribution. Let B_c denote the selection budget allocated to class c, and let T_c denote the set of target-slice source examples in that class. The deterministic enrichment result then guarantees at least $\min B_c, |T_c|$ selected target-slice examples within class c. Under the appendix's repair-utility condition, this additional target support yields expected repair utility at least as high as random or stratified selection with the same class allocation, with a strict advantage when the enrichment gap is strict. `The selected subset can therefore be used for checkpoint fine-tuning, continued training, or retraining from scratch.` The update rule determines how the additional target support affects the final metric; it does not change the selection guarantee. We do not claim final-metric superiority for every repair pipeline, generator, optimizer, or fine-tuning strategy.

5. Attribute-Generation Ablation

We reran the source-only evaluation with attributes from Qwen3-VL, BLIP, and InternVL3, using the same five seeds in each case.

Five-seed ImageNet10 test accuracy (%). `One-Shot Mean` averages source ratios `0.1/0.3/0.5/0.7/0.9`; `AL Curve Mean` averages labeled ratios `0.1--0.6`; and `Final ACC` is the Active Learning accuracy at ratio `0.6`.

Attribute Source	One-Shot Mean	AL Curve Mean	Final ACC at Ratio 0.6
Qwen3-VL (ours)	62.96	**58.09**	**68.80**
BLIP	62.18	55.99	65.46
InternVL3	**63.27**	56.33	66.23

`No attribute source is strongest in both settings.` InternVL3 has the highest One-Shot Mean, 0.31 accuracy points above Qwen3-VL. Qwen3-VL performs best in Active Learning, exceeding InternVL3 by 1.76 points in Curve Mean and 2.57 points at ratio 0.6. AttriSelect works with all three attribute sources, but the choice of source affects the result.

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Replying to Response to Area Chair 6k27 (Part 4)

Response to Area Chair 6k27 (Part 5)

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Official Comment by Authors 30 Jul 2026, 02:54

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6. Other Pipeline Components

The submitted paper ablates all three AttriSelect score components: class-conditional rarity (`R`), the generated-data frequency penalty (`P`), and soft class balancing (`C`).

ImageNet10 component ablation, test accuracy.

Ratio	None	R	P	C	P+C	R+C	R+P	R+P+C
0.1	0.7269	0.7300	0.7262	0.7277	0.7359	0.7308	0.7315	**0.7369**
0.3	0.7308	0.7415	0.7392	0.7392	0.7308	0.7369	0.7315	**0.7492**
0.5	0.7400	0.7415	0.7408	0.7346	0.7369	0.7415	0.7446	**0.7508**

Ratio	None	R	P	C	P+C	R+C	R+P	R+P+C
0.7	0.7423	0.7469	0.7431	0.7415	0.7431	0.7469	0.7477	**0.7515**
0.9	0.7462	0.7500	0.7526	0.7477	0.7508	0.7415	0.7423	**0.7562**

`The full R+P+C score is strongest at all five ImageNet10 source ratios.` Removing `R`, `P`, or `C` reduces mean test accuracy by `0.00942`, `0.00940`, or `0.00940`, respectively. The single-component variants are also weaker on average (`R: 0.74198`, `P: 0.74038`, and `C: 0.73814`) than the full score (`0.74892`). The nearly identical leave-one-out drops suggest that the components work together; none is dominant on ImageNet10.

COCO component ablation, keypoint AP. The corresponding study is reported in the appendix.

Ratio	None	R	P	C	P+C	R+C	R+P	R+P+C
0.1	0.6048	0.6099	0.5983	0.6067	0.6066	0.6149	0.5954	**0.6183**
0.3	0.6040	0.6134	0.6038	0.6136	0.6142	0.6196	0.6136	**0.6306**
0.5	0.6209	0.6074	0.6113	0.6149	0.6295	0.6246	0.6155	**0.6317**
0.7	0.6151	0.6222	0.6231	**0.6319**	0.6317	0.6302	0.6282	0.6311
0.9	0.6091	0.6175	0.6174	0.6270	0.6276	0.6225	0.6231	**0.6355**

`The full score is strongest at four of five COCO ratios and remains within 0.0008 AP of the best result at ratio 0.7.` Removing `R`, `P`, or `C` reduces mean AP by `0.00752`, `0.00708`, or `0.01428`, respectively. Single-component variants are less reliable across source ratios. On both datasets, the full score benefits from combining rarity, generated-data frequency control, and soft class balancing. The appendix also contains the alternative-generator experiments, the VLM human audit, and the attribute-noise sensitivity study.

7. Incomplete Attribute-Vocabulary Coverage

To test incomplete vocabulary coverage, we remove one target field while holding the generated repair pool, source budget, checkpoint, and training schedule fixed. Parentheses give the selected-set Jaccard overlap with the full two-field selector.

ImageNet10: `pink AND fabric`, overall accuracy (%).

Ratio	Full	Missing Color	Missing Texture
0.1	73.69	72.69 (0.5604)	**73.85** (0.3904)
0.3	**74.92**	73.77 (0.6166)	73.54 (0.5842)
0.5	**75.08**	74.23 (0.6443)	74.23 (0.7330)
0.7	**75.15**	74.00 (0.6999)	74.23 (0.8826)
0.9	**75.62**	75.08 (0.8765)	74.54 (0.9668)

COCO: `brown AND rainy`, keypoint AP.

Ratio	Full	Missing Color	Missing Weather
0.1	**0.6183**	0.6055 (0.4782)	0.6105 (0.2034)
0.3	**0.6306**	0.6221 (0.5611)	0.6214 (0.4613)
0.5	**0.6317**	0.6178 (0.7063)	0.6271 (0.6084)
0.7	0.6319	**0.6332** (0.7836)	0.6323 (0.8110)
0.9	**0.6355**	0.6237 (0.8969)	0.6244 (0.8708)

> **Vocabulary-coverage takeaway.** A missing target field changes the selected source set most at low budgets. At ratio `0.1`, Jaccard falls to `0.5604/0.3904` for missing color/texture on ImageNet10 and `0.4782/0.2034` for missing color/weather on COCO. The downstream effect is not monotonic. Mean ImageNet10 accuracy changes are `−0.94/−0.81` points for missing color/texture, while mean COCO AP changes are `−0.0091/−0.0065` for missing color/weather. `Missing-field variants are lower than Full in 17 of the 20 dataset--field--ratio comparisons.` Incomplete coverage therefore changes source selection and usually lowers task performance, but not at every ratio.

References

1. SafeFix: Targeted Model Repair via Controlled Image Generation. TMLR, 2026.
2. Actively Selecting Annotations Among Objects and Attributes. ICCV, 2011.
3. HiBug: On Human-Interpretable Model Debug. NeurIPS, 2023.

Add: **Official Comment** **Author AC Confidential Comment**

Official Review of Submission12417 by Reviewer yyPZ

Official Review by Reviewer yyPZ 25 Jun 2026, 01:34 (modified: 28 Jul 2026, 13:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=gFhG8tCROU)

Summary:
The paper introduces AttriSelect, a targeted data selection algorithm designed to efficiently repair semantic failure slices in computer vision models. Rather than relying on traditional active learning metrics like uncertainty, gradient embeddings, or geometric coverage, AttriSelect scores source training data using structural attribute metadata assigned by Vision-Language Models (VLMs).

The scoring function balances three components: class-conditional attribute rarity, a penalty for attribute values already abundant in the generated synthetic repair data, and a soft class-balancing term.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths:

1. Framing targeted model repair as an attribute-aware source selection problem is a highly practical perspective.
2. The methodology is straightforward.
3. Writing is clear.

Weakness:

1. Misleading Claims Regarding Computational Efficiency. The authors highlight in Table 2 that AttriSelect takes less than 2 seconds, claiming a speedup of up to 283× over baselines. However, this entirely omits the massive computational overhead of the prerequisite step: running Qwen3-VL to annotate the entire source pool. Appendix A states it takes 10–20 minutes to annotate just 1,000 images. For datasets like CelebA (80,000 images) or COCO (20,000 images), this requires tens to hundreds of GPU hours. Claiming high efficiency while ignoring the cost of generating the metadata that the algorithm relies upon is highly misleading.
2. Comparison look Unfair. AttriSelect is given privileged information like VLM-assigned attributes and explicit bug definitions, but the compared active learning baselines (Entropy, BADGE, CoreSet) do not have access to. The authors should compare it against a baseline with the same information, for example, an "Attribute-Filtered Random" or "Attribute-Filtered Entropy" baseline.
3. Despite the heavy pipeline (VLM tagging, diffusion generation, subset selection), the absolute Fix Rates (FR) in Table 1 remain quite low, mostly ranging between 5% and 13%.

Quality: 2: not good
Clarity: 3: good
Significance: 2: not good
Originality: 3: good
Questions:
See weaknesses

Limitations:
yes

Rating: 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility and incompletely addressed ethical considerations.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
NA

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer yyPZ (Part 3)

Official Comment
by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))
29 Jul 2026, 22:03 (modified: 29 Jul 2026, 22:03)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ
Revisions (/revisions?id=hRlVhQ5ggp)

[Deleted]

Response to Reviewer yyPZ (Part 1)

Edit

Official Comment
by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))
29 Jul 2026, 22:01
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:
Dear Reviewer yyPZ, thank you for raising these questions about end-to-end cost, comparison fairness, and the interpretation of FR.

This response is organized into four parts.

Q1 — End-to-End Computational Cost

Accounting Scope

Response. Table 2 of the submitted paper measures only source-subset selection time, not the complete repair pipeline, and the revision will make that scope explicit. The reported speedups apply to source selection after the attribute records are available. **They should not be interpreted as end-to-end speedups** over source annotation, repair-data construction, retraining, and evaluation.

We report repair-image generation separately because **the generated repair set is a fixed, publicly available dataset shared by every selector**, not a set regenerated for each method. Every selector receives the same repair images as pre-existing augmentation data. Regenerating the SafeFix [1] set would add the same one-time dataset-construction cost to every method without changing their runtime ordering. The table columns therefore cover source-side preprocessing and selection through retraining and evaluation. By contrast, **we include VLM source-pool assignment for AttriSelect** because it is a method-specific prerequisite.

Measured Source-Pool Assignment Time

To answer the assignment-cost question directly, we measured complete source-pool VLM assignment on one NVIDIA H200 GPU:

Dataset	Source Images	Assigned Attributes	H200 Assignment Time	Normalized Assignment Time (min / 1K images / attribute)
KITTI	5,000	6	32.77 min	1.09
ImageNet10	10,400	3	29.35 min	0.94
COCO	20,000	5	90 min	0.90

`Normalized Assignment Time` divides the total H200 assignment time by the source-pool size in thousands and the number of assigned attributes. For example, the COCO value is `90 / (20 x 5) = 0.90` minutes per 1,000 images per attribute.

Conservative Comparison

For a fixed task-level attribute vocabulary, **source-pool assignment is a one-time preprocessing cost, and its metadata can be reused** across source ratios, selectors, and later repair iterations on the same task. The full pool does not need to be annotated again for every repair. Here, **the first repair run assumes that source attributes must be assigned from scratch**. Its cost covers source preprocessing and selection through retraining and evaluation; repair-image generation is reported separately. Even so, **every AttriSelect row below charges the full fresh-assignment cost**. On one NVIDIA H200 GPU, the mean over two complete 5,000-image KITTI runs is **32.77 minutes**. The earlier estimate of 10--20 minutes per 1,000 images in the submitted paper was measured on an NVIDIA A5000, as stated in the appendix; on the H200, assignment takes several minutes per 1,000 images, depending on the number of required attributes. Assigning the complete 20,000-image COCO source pool on the same H200 takes approximately **1.50 hours**.

KITTI: 5,000 source images. Each AttriSelect row charges the mean fresh 5,000-image H200 VLM assignment time of 32.77 minutes.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	42.17 min	42.18 min	48.82 min	1.37 h	1.12 h	43.47 min	43.53 min	1.01 h
0.3	43.84 min	43.85 min	50.28 min	1.44 h	1.20 h	45.08 min	45.63 min	1.09 h
0.5	45.18 min	45.18 min	51.73 min	1.49 h	1.27 h	46.42 min	47.47 min	1.11 h
0.7	47.04 min	47.05 min	53.46 min	1.56 h	1.35 h	48.16 min	49.73 min	1.13 h
0.9	48.86 min	48.87 min	55.53 min	1.63 h	1.42 h	50.34 min	51.46 min	1.15 h

COCO: 20,000 source images. Image generation and generated-repair-image assignment are excluded. Each entry includes the applicable source preprocessing and selection cost, retraining, and evaluation. The AttriSelect column charges a fresh full-pool VLM assignment in every row.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	4.41 min	4.42 min	10.98 min	16.18 min	19.63 min	5.08 min	6.97 min	1.57 h

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.3	8.18 min	8.18 min	14.74 min	39.86 min	49.76 min	8.93 min	17.86 min	1.62 h
0.5	12.37 min	12.36 min	19.04 min	1.08 h	1.56 h	13.08 min	29.47 min	1.66 h
0.7	15.73 min	15.75 min	22.27 min	1.45 h	2.29 h	16.34 min	37.63 min	1.69 h
0.9	19.66 min	19.65 min	26.28 min	1.87 h	3.01 h	20.18 min	40.94 min	1.72 h

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer yyPZ (Part 2)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:02

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Q1 — End-to-End Computational Cost (continued)

Cost–Performance Interpretation

> **Cost takeaway.** Combined with the main results in the submitted paper, these tables show a trade-off between computational cost and repair performance. **Random, Stratified, UHerding, and PPAL are faster, especially at small source ratios, but their repair performance is generally lower.** For example, at COCO ratio 0.1, AttriSelect takes 1.57 hours when fresh annotation is charged, whereas Random, UHerding, and PPAL take 4.41, 5.08, and 6.97 minutes, respectively; their test AP values are 0.6048, 0.6013, and 0.5950, compared with 0.6183 for AttriSelect.
> **CoreSet and BADGE are generally the two strongest baselines besides AttriSelect, but they become much slower at high source ratios.** At ratio 0.9, they are even slower than AttriSelect after the AttriSelect total conservatively includes VLM annotation of the entire source pool. On COCO, AttriSelect takes 1.72 hours and reaches 0.6355 AP, compared with 1.87 hours/0.6202 for CoreSet and 3.01 hours/0.6213 for BADGE. On KITTI, AttriSelect takes 1.15 hours and reaches 0.5850 mAP50--95, compared with 1.63 hours/0.5781 for CoreSet and 1.42 hours/0.5810 for BADGE. Thus, **at ratio 0.9 on both COCO and KITTI, AttriSelect provides both better repair performance and lower wall-clock cost on the first repair run, even after charging full-pool VLM assignment**.

CoreSet [3] and BADGE [4] become substantially slower at high source ratios and on larger source pools because their standard greedy procedures update the distance from every one of the N candidates to each new center. Selecting $B = rN$ samples requires $O(NB) = O(rN^2)$ distance-update work at a fixed ratio. The measurements follow this scaling pattern: cost rises sharply with the retained ratio and grows further with the source pool. BADGE can be slower because its gradient embeddings are usually higher-dimensional. AttriSelect does not repeat candidate-to-center computations after its reusable attribute records are available.

Existing Metadata and Later Repair Cycles

Source attributes also need not be assigned from scratch when suitable metadata already exist. For example, the official CelebA release (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html)) provides 40 binary attributes for each of its 202,599 images. **If these public annotations cover the required task-level vocabulary, AttriSelect can use them directly and incurs no VLM assignment cost.** New assignment is needed only for attributes missing from the available metadata.

Cost After Metadata Reuse

The tables are deliberately conservative for AttriSelect because they charge source assignment separately at every ratio. **In repeated repair cycles, the metadata are reused, so the annotation cost is amortized and the remaining online AttriSelect computation is below two seconds.** We will revise Table 2 to label its values explicitly as source-subset selector runtime, add this end-to-end decomposition, and clarify that **the reported 39--283x speedups apply only to source-subset selection after attribute data are available**.

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer yyPZ (Part 3)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:04 (modified: 29 Jul 2026, 22:09)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=r4UQKe5wbF)

Comment:

Q2 — Same-Information Attribute-Aware Baselines

Response. The generic active-learning baselines do not receive the attribute information used by AttriSelect. They test whether repair-specific metadata adds signal beyond uncertainty, geometric coverage, and gradient embeddings, but they do not test whether our score uses that metadata better than a simple attribute-aware rule.

On ImageNet10, the target slice is defined by `color=pink` and `texture=fabric`. We added three baselines that receive the same attribute metadata and target definition:

- **Target-Slice + Stratified:** select exact `pink`-and-`fabric` source examples first and fill the remaining budget with class-stratified background samples;
- **Attribute-Filtered Random:** prioritize the `pink`-or-`fabric` source pool and sample randomly within class-balanced queues before filling from the background;
- **Attribute-Filtered Entropy:** use the same candidate pool and class-balanced queues, but rank candidates by predictive entropy from the fixed diagnosis checkpoint.

All methods select only source images and use the same fixed 1,000-image generated set, source budget, VLM annotations, class labels, diagnosis checkpoint, generated seed, and 15-epoch repair schedule. The table reports mean overall ACC and standard deviation over seeds 0, 42, 1234, 3407, and 2024. The AttriSelect values are the five-seed results reported in the appendix of the submitted paper.

Five-seed ImageNet10 same-information repair comparison. Metric: test accuracy (fraction; mean ± standard deviation).

Source ratio	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.1	0.7303 ± 0.0051	0.7340 ± 0.0034	0.7317 ± 0.0025	**0.7369 ± 0.0043**
0.3	0.7385 ± 0.0024	0.7382 ± 0.0042	0.7365 ± 0.0023	**0.7492 ± 0.0044**
0.5	0.7438 ± 0.0037	0.7442 ± 0.0038	0.7442 ± 0.0031	**0.7508 ± 0.0010**
0.7	0.7434 ± 0.0065	0.7474 ± 0.0038	0.7482 ± 0.0020	**0.7515 ± 0.0037**
0.9	0.7485 ± 0.0026	0.7469 ± 0.0036	0.7503 ± 0.0045	**0.7562 ± 0.0019**

> **Key result.** Even when every baseline receives the same attribute metadata and target-slice definition, AttriSelect has the highest mean accuracy at all five source ratios. Its margin over the strongest same-information baseline ranges from `0.29` to `1.07` accuracy points, with an average of `0.59` points. This consistent but moderate advantage shows that the selection score provides additional value beyond access to the attributes alone.

Q3 — Interpreting Fix Rate and the Repair Budget

> “Despite the heavy pipeline (VLM tagging, diffusion generation, subset selection), the absolute Fix Rates (FR) in Table 1 remain quite low, mostly ranging between 5% and 13%.”

Raw-Metric Interpretation

Response. Fix Rate (FR) should be read together with the original task metric. FR is the **Relative Error Reduction** toward the maximum task score, not the percentage of test examples or failure slices fixed. At the reported ratio-0.9 AttriSelect settings, the raw changes are:

- ImageNet10 accuracy: **0.7173 to 0.7562, or +3.89 percentage points**;
- CelebA accuracy: **0.9057 to 0.9196, or +1.39 percentage points**;
- COCO keypoint AP: **0.5902 to 0.6355, or +4.53 AP**;
- KITTI mAP50--95: **0.5626 to 0.5850, or +2.24 mAP**.

The slice-level heatmaps in Figures 2 and 4 of the submitted paper provide a more direct view of the localized repair effect. At source ratio `0.5`, the largest improvements occur on the diagnosed values---pink and fabric for ImageNet10, and brown and rainy for COCO---while the remaining attribute slices show only small changes and no systematic degradation. Thus, the overall metric can understate the targeted benefit: AttriSelect improves the diagnosed failure slices without materially reducing performance on unrelated slices.

The reported 5--13% range represents a reduction in the model's remaining error, not a 5--13 percentage-point change in the raw task metric. AttriSelect already achieves **6.93% Relative Error Reduction at ImageNet10 ratio 0.1**, raising ACC from 0.7173 to 0.7369 (+1.96 percentage points) while selecting only 10% of the source pool. This is the highest FR at ratio 0.1: the strongest baseline, UHerding, reaches 5.59% FR and 0.7331 ACC. AttriSelect improves FR by **1.34 points (24.0% relative)** and ACC by **0.38 percentage points** under the same budget. Ratio 0.1 is also the most data- and retraining-efficient repair setting evaluated: it uses one-ninth as many selected source examples as ratio 0.9 while retaining the same generated repair set. The fixed source annotation and image-generation costs are accounted for separately in the end-to-end analysis.

Add: **Official Comment** **Author AC Confidential Comment**

Replying to Response to Reviewer yyPZ (Part 3)

Response to Reviewer yyPZ (Part 4)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:05

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Q3 — Interpreting Fix Rate and the Repair Budget (continued)

Epoch-Budget Sensitivity

As specified in the appendix, the ImageNet10 diagnosis model is trained for 30 epochs at learning rate 1×10^{-4}, while repair uses 15 epochs at 1×10^{-5}. For CelebA, COCO, and KITTI, repair likewise **uses half as many epochs as diagnosis**, with learning rate 1×10^{-5}. **This moderate schedule makes the experiment a controlled comparison of source-selection methods within model repair. Because AttriSelect is a data-selection method, we compare it with different AL baselines under a matched repair budget instead of prolonging training to maximize performance on the underlying computer vision task.** Every selector uses the same number of repair epochs, learning rate, generated set, and source-data budget.

To test whether the reported FR values are mainly limited by this schedule, we also vary ImageNet10 repair training from 15 to 60 epochs, corresponding to $0.5\times$--$2\times$ the 30-epoch diagnosis schedule. We evaluate source ratios 0.5 and 0.9 and average over the same five seeds (0, 42, 1234, 3407, and 2024) used in the main paper.

ImageNet10, source ratio 0.5 (FR, %).

Method	**15 epochs ($0.5\times$)**	**30 epochs ($1.0\times$)**	**45 epochs ($1.5\times$)**	**60 epochs ($2.0\times$)**
Random	7.32	9.04	10.04	10.27
Stratified	9.66	9.44	10.49	11.06
Entropy	5.59	10.57	11.08	11.67
CoreSet	8.03	9.99	10.21	11.56
BADGE	10.47	10.74	11.84	12.11
UHerding	8.38	8.94	9.31	10.17
PPAL	8.84	9.29	10.34	10.56
AttriSelect	**11.85**	**12.39**	**13.51**	**14.08**

ImageNet10, source ratio 0.9 (FR, %).

Method	**15 epochs ($0.5\times$)**	**30 epochs ($1.0\times$)**	**45 epochs ($1.5\times$)**	**60 epochs ($2.0\times$)**
Random	10.22	11.02	12.15	11.53
Stratified	10.47	12.05	12.91	13.22
Entropy	10.22	12.33	13.44	13.76

Method	15 epochs ($0.5\times$)	30 epochs ($1.0\times$)	45 epochs ($1.5\times$)	60 epochs ($2.0\times$)
CoreSet	6.12	12.08	12.31	13.39
BADGE	9.37	10.05	11.09	11.48
UHerding	11.57	12.11	12.37	12.60
PPAL	11.85	12.19	12.81	13.01
AttriSelect	**13.76**	**14.25**	**14.99**	**15.22**

Extending repair training from 15 to 60 epochs raises AttriSelect ACC from 0.7508 to 0.7571 at ratio 0.5 and from 0.7562 to 0.7603 at ratio 0.9. Relative to the pre-repair ACC of 0.7173, the 60-epoch results correspond to gains of 3.98 and 4.30 percentage points, respectively. Over the same schedules, AttriSelect FR increases from 11.85% to 14.08% and from 13.76% to 15.22%. Most baselines also improve overall, although individual trajectories are not strictly monotonic.

> **Key result.** AttriSelect is best in **all eight ratio--epoch configurations**, leading the strongest baseline by 1.38--1.97 FR points at ratio 0.5 and 1.46--1.92 points at ratio 0.9. The persistent ordering supports a selected-set quality effect rather than an early-stopping artifact.

The generation and diagnosis stages are held fixed across source-selection methods, so Table 1 of the submitted paper and this additional experiment isolate which source subset gives the best repair result under the same upstream repair data and source budget. We will revise the main results to show raw metrics and absolute changes next to FR and state its interpretation more precisely.

References

1. SafeFix: Targeted Model Repair via Controlled Image Generation. TMLR, 2026.
2. Actively Selecting Annotations Among Objects and Attributes. ICCV, 2011.
3. Active Learning for Convolutional Neural Networks: A Core-Set Approach. ICLR, 2018.
4. Deep Batch Active Learning by Diverse, Uncertain Gradient Lower Bounds. ICLR, 2020.

Add: **Official Comment** **Author AC Confidential Comment**

Official Review of Submission12417 by Reviewer hG6j

Official Review by Reviewer hG6j 19 Jun 2026, 05:39 (modified: 28 Jul 2026, 13:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=qvZIzZToTG)

Summary:
This paper studies efficient source-data selection for targeted vision model repair. Given an already diagnosed failure slice and a fixed set of generated repair images, the goal is to choose a compact subset of the original source data for retraining. The proposed method, AttriSelect, scores source examples using class-conditional attribute rarity, a generated-data frequency penalty, and soft class balancing. Experiments on classification, pose estimation, and detection show that AttriSelect often outperforms generic active learning baselines and can sometimes outperform full-source repair.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:
Strength: The paper addresses a useful and relatively underexplored problem: after a bug slice has been found and repair images have been generated, which original source examples should be used for efficient repair? This is a practical question that is not well covered by standard active learning. The method is simple, intuitive, and well aligned with the problem. The three scoring terms each have a clear role: rarity targets rare semantic attributes, the

generated-data penalty avoids redundancy with synthetic repair data, and soft class balancing avoids class collapse. The experiments are reasonably broad and include several strong active learning baselines. The runtime advantage is also convincing.

Weakness The main weakness is that the method depends heavily on the upstream slice discovery, generated data quality, and VLM-assigned attributes. AttriSelect does not itself discover the failure slice, and the paper could be clearer about how robust it is to different slice definitions or imperfect attribute vocabularies. The algorithmic novelty is also moderate: the score is useful but fairly straightforward. I would also like to see stronger attribute-aware baselines, such as selecting target-slice examples plus stratified background samples.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:

1. How sensitive is AttriSelect to the upstream slice discovery method and slice granularity?
2. Can the authors compare with simple attribute-aware baselines, such as exact target-slice selection plus stratified sampling?
3. What is the end-to-end cost including generation, VLM attribute assignment, selection, and retraining?
4. How does the method behave when the failure slice is not well captured by the closed-set attribute vocabulary?

Limitations:
The paper discusses attribute noise and includes a human audit, which is helpful. However, it should more explicitly discuss dependence on upstream slice discovery, VLM attribute quality, generator quality, and end-to-end repair cost.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
No

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer hG6j (Part 1)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:11

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Comment:
Dear Reviewer hG6j, thank you for the questions about the upstream interface, slice granularity, comparison baselines, cost, and vocabulary coverage.

This response is organized into four parts.

Q1 — Upstream Slice Discovery and Granularity

> “How sensitive is AttriSelect to the upstream slice discovery method and slice granularity?”

Upstream Interface

Response. AttriSelect starts after an upstream debugging method has diagnosed and semantically described a failure slice. As specified in the appendix, the attribute vocabulary, target fields, and VLM assignment prompt are fixed before source selection. Even when the same VLM family contributes to target-slice wording and image-level attribute assignment, the two roles are separated: the slice description is fixed first, and assignment then runs as a frozen closed-set classifier over source and generated images. The assignment stage does not receive model predictions, failure rates, selector identities, selected-set membership, or downstream repair results.

Under this interface, **two discovery methods that produce the same canonical target conjunction and generated repair set give AttriSelect the same inputs and therefore the same selection result**. The discovery algorithm matters only if it changes the downstream semantic slice, including its fields, target values, granularity, or generated repair data. We will state this condition explicitly: **AttriSelect does not discover or validate the failure slice itself**; it selects source data for an existing debugging and generation pipeline.

Annotation Reliability

The appendix quantifies the reliability of this semantic interface through a blinded human audit. Each of five AI graduate students labels 100 CelebA, 100 COCO, and 100 KITTI images using the same closed-set options as the VLM. The annotators do not see VLM labels, target flags, model predictions, data splits, or method outputs. Majority-vote human labels are then compared with the VLM assignments:

Dataset	Target field 1 agreement	Target field 2 agreement	Slice-indicator agreement
CelebA	Hair color: 98%	Skin tone: 99%	92%
COCO	Clothes color: 97%	Weather: 91%	90%
KITTI	Object color: 98%	Weather: 96%	94%

Slice-indicator agreement is the most relevant measure because AttriSelect uses the target conjunction, not either field in isolation. **Agreement is 90--94% across the three audited datasets**, so the selector's semantic slices largely match independent human judgments.

The appendix also corrupts the target-field assignments while keeping the generated repair set, source ratio, and training recipe fixed. At source ratio 0.5, the changes from the clean-VLM AttriSelect result are:

Dataset	Task metric	5% attribute noise	10% attribute noise	20% attribute noise
CelebA	Test ACC change	0.007	0.01	0.09
COCO	Test AP change	0.002	0.003	0.02
KITTI	Test mAP50--95 change	0.003	0.004	0.06

The small changes under 5% and 10% corruption show that a few isolated assignment errors do not drive AttriSelect. At 20% noise, the larger changes mark the expected failure regime: the attribute counts no longer represent the diagnosed slice reliably. The audit and corruption experiments test the fixed semantic interface. The next experiment instead changes the slice definition.

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer hG6j (Part 2)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:11

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Comment:

Q1 — Upstream Slice Discovery and Granularity (continued)

Slice-Granularity Sensitivity

Slice granularity can still affect repair because it changes both the semantic factors used for source selection and the generated repair set. A coarse slice can mix different failure causes and dilute the relevant attribute signal. A narrow conjunction may have too little support and exclude useful examples that match the main failure factors but not every added condition.

We evaluate this end-to-end sensitivity on COCO and ImageNet10 over five seeds and source ratios 0.1--0.9. For COCO, we compare the diagnosed two-attribute slice `brown AND rainy` with Brown-only, Rain-only, and the narrower three-attribute slice `brown AND rainy AND night`:

Source ratio	Brown-only AP	Rain-only AP	Brown + rainy + night AP	Brown + rainy AP(Ours)
0.1	0.6161	0.6174	0.6097	**0.6183**
0.3	0.6238	0.6224	0.6179	**0.6306**
0.5	0.6284	0.6217	0.6246	**0.6317**
0.7	0.6327	**0.6333**	0.6291	0.6319
0.9	0.6348	0.6132	0.6336	**0.6355**

The diagnosed two-attribute slice exceeds Brown-only and Rain-only at four of five source ratios, with average gains of **0.24 and 0.80 AP points**, respectively. It exceeds the narrower three-attribute slice at **all five ratios by 0.66 AP points on average**.

For ImageNet10, we compare the diagnosed `pink AND fabric` slice with Pink-only, Fabric-only, and the narrower `pink AND fabric AND rainy background` slice:

Source ratio	Pink-only ACC	Fabric-only ACC	Pink + fabric + rainy ACC	Pink + fabric ACC
0.1	0.7223	0.7346	**0.7398**	0.7369
0.3	0.7369	0.7431	0.7414	**0.7492**
0.5	0.7377	0.7438	0.7429	**0.7508**
0.7	0.7485	0.7423	0.7462	**0.7515**
0.9	0.7500	0.7354	0.7491	**0.7562**

The diagnosed two-attribute slice exceeds both single-attribute variants at all five source ratios, by **0.98 and 0.91 accuracy points on average**, and exceeds the narrower three-attribute slice at four of five ratios by **0.50 points on average**.

> **Granularity takeaway. Downstream performance is relatively insensitive to moderate changes in slice granularity**, although **granularity still matters**: the diagnosed two-attribute definition performs best in **8 of the 10 dataset-ratio settings**. This pattern suggests a support--specificity trade-off: a coarse slice has greater support but includes fewer relevant examples, whereas an overly narrow conjunction can exclude useful examples.

Q2 — Simple Attribute-Aware Baselines

Response. On ImageNet10, the target slice is defined by `color=pink` and `texture=fabric`. We implemented exact target-slice selection with stratified background sampling, along with two additional attribute-aware controls:

- **Target-Slice + Stratified:** select exact `pink`-and-`fabric` source examples first and fill the remaining budget with class-stratified background samples;
- **Attribute-Filtered Random:** prioritize the `pink`-or-`fabric` source pool and sample randomly within class-balanced queues before filling from the background;
- **Attribute-Filtered Entropy:** use the same candidate pool and class-balanced queues, but rank candidates by predictive entropy from the fixed diagnosis checkpoint.

All methods select only source images and use the same fixed 1,000-image generated set, source budget, VLM annotations, class labels, diagnosis checkpoint, generated seed, and 15-epoch repair schedule. The table reports mean overall ACC and standard deviation over seeds 0, 42, 1234, 3407, and 2024. The AttriSelect values are the five-seed results reported in the appendix of the submitted paper.

Source ratio	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.1	0.7303 ± 0.0051	0.7340 ± 0.0034	0.7317 ± 0.0025	**0.7369 ± 0.0043**
0.3	0.7385 ± 0.0024	0.7382 ± 0.0042	0.7365 ± 0.0023	**0.7492 ± 0.0044**
0.5	0.7438 ± 0.0037	0.7442 ± 0.0038	0.7442 ± 0.0031	**0.7508 ± 0.0010**
0.7	0.7434 ± 0.0065	0.7474 ± 0.0038	0.7482 ± 0.0020	**0.7515 ± 0.0037**
0.9	0.7485 ± 0.0026	0.7469 ± 0.0036	0.7503 ± 0.0045	**0.7562 ± 0.0019**

> **Key result.** Even when every baseline receives the same attribute metadata and target-slice definition, AttriSelect has the highest mean accuracy at all five source ratios. Its margin over the strongest same-information baseline ranges from `0.29` to `1.07` accuracy points, with an average of `0.59` points. This consistent but moderate advantage shows that the selection score provides additional value beyond access to the attributes alone.

Add: **Official Comment** **Author AC Confidential Comment**

Replying to Response to Reviewer hG6j (Part 2)

Response to Reviewer hG6j (Part 3)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:12

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Comment:

Q3 — End-to-End Cost Accounting

> "What is the end-to-end cost including generation, VLM attribute assignment, selection, and retraining?"

Accounting Scope

Response. Table 2 of the submitted paper measures only source-subset selection time, not the complete repair pipeline, and the revision will make that scope explicit. The reported speedups apply to source selection after the attribute records are available. **They should not be interpreted as end-to-end speedups** over source annotation, repair-data construction, retraining, and evaluation.

Repair-image generation is a fixed upstream SafeFix [1] dataset-construction stage shared by every selector. Every method receives the same publicly available repair images, so regenerating them from scratch would add the same cost to every method without changing their runtime ordering. We therefore report generation separately rather than charging it only to AttriSelect. In contrast, **we include full source-pool VLM assignment for AttriSelect** because it is a method-specific prerequisite.

Conservative Cost of the First Repair Run

Source-pool assignment is a one-time preprocessing cost for a fixed task-level vocabulary. Its metadata can be reused across source ratios and later repair cycles. Here, **the first repair run assumes that source attributes must be assigned from scratch**. Its cost covers source preprocessing and selection through retraining and evaluation; repair-image generation is reported separately. The tables conservatively charge a fresh full-pool assignment in every AttriSelect row. For KITTI, the mean measured assignment time over the two complete 5,000-image runs is **32.77 minutes** on a single NVIDIA H200 GPU (the previously reported 10--20 minutes per 1,000 images was measured on an NVIDIA A5000, whereas an H200 requires only approximately 5 minutes per 1,000 images). On the same GPU, assigning attributes to the complete 20,000-image COCO source pool takes approximately **1.50 hours**.

KITTI: 5,000 source images. Each AttriSelect row charges the mean fresh 5,000-image H200 VLM assignment time of 32.77 minutes.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	42.17 min	42.18 min	48.82 min	1.37 h	1.12 h	43.47 min	43.53 min	1.01 h
0.3	43.84 min	43.85 min	50.28 min	1.44 h	1.20 h	45.08 min	45.63 min	1.09 h
0.5	45.18 min	45.18 min	51.73 min	1.49 h	1.27 h	46.42 min	47.47 min	1.11 h
0.7	47.04 min	47.05 min	53.46 min	1.56 h	1.35 h	48.16 min	49.73 min	1.13 h
0.9	48.86 min	48.87 min	55.53 min	1.63 h	1.42 h	50.34 min	51.46 min	1.15 h

COCO: 20,000 source images. Image generation and generated-repair-image assignment are excluded. Each entry includes the applicable source preprocessing and selection cost, retraining, and evaluation. The AttriSelect column charges a fresh full-pool VLM assignment in every row.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	4.41 min	4.42 min	10.98 min	16.18 min	19.63 min	5.08 min	6.97 min	1.57 h
0.3	8.18 min	8.18 min	14.74 min	39.86 min	49.76 min	8.93 min	17.86 min	1.62 h
0.5	12.37 min	12.36 min	19.04 min	1.08 h	1.56 h	13.08 min	29.47 min	1.66 h
0.7	15.73 min	15.75 min	22.27 min	1.45 h	2.29 h	16.34 min	37.63 min	1.69 h
0.9	19.66 min	19.65 min	26.28 min	1.87 h	3.01 h	20.18 min	40.94 min	1.72 h

> **Cost takeaway.** At small source ratios, lightweight conventional selectors have lower wall-clock cost on the first repair run. At ratio 0.9 on both COCO and KITTI, however, **AttriSelect is better than CoreSet and BADGE in both repair performance and first-run wall-clock cost**.

On COCO, AttriSelect takes 1.72 hours and reaches 0.6355 AP, compared with 1.87 hours/0.6202 for CoreSet and 3.01 hours/0.6213 for BADGE. On KITTI, it takes 1.15 hours and reaches 0.5850 mAP50--95, compared with 1.63 hours/0.5781 and 1.42 hours/0.5810.

Add: **Official Comment** **Author AC Confidential Comment**

Replying to Response to Reviewer hG6j (Part 3)

Response to Reviewer hG6j (Part 4)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:13

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Comment:

Q3 — End-to-End Cost Accounting (continued)

Existing Metadata and Later Repair Cycles

Source attributes need not be assigned from scratch when suitable metadata already exist. For example, the official CelebA release (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html)) provides 40 binary attributes for each of its 202,599 images. **If these public annotations cover the required task-level vocabulary, AttriSelect can use them directly and incurs no VLM assignment cost.** New assignment is needed only for attributes missing from the available metadata.

Cost After Metadata Reuse

The tables are deliberately conservative because they charge source assignment again at every ratio. **In repeated repair cycles, the metadata are reused, so this annotation cost is amortized and the remaining online AttriSelect computation is below two seconds.** We will revise Table 2 to label its values explicitly as source-subset selector runtime and distinguish one-time preprocessing, the first repair run, and later runs with reused metadata.

Q4 — Incomplete Attribute-Vocabulary Coverage

> "How does the method behave when the failure slice is not well captured by the closed-set attribute vocabulary?"

Response. This question concerns coverage of the attribute schema, not synonymous wording. Once names and values are mapped to canonical closed-set labels, changing their surface form does not change the AttriSelect score. The two substantive failure cases are a missing factor and a category that merges distinct values too coarsely.

If a target factor is absent, AttriSelect cannot explicitly prioritize that factor. It can use only the remaining observed attributes and their correlations with the failure slice. The repair-specific part of the score therefore weakens, while soft class balance still preserves label coverage. **If target and non-target values are merged, class-conditional rarity and the generated-data frequency penalty become less discriminative**, which can cause the selected subset to include more examples outside the true fine-grained slice. These are limitations of the current closed-set formulation.

We ran controlled missing-field studies on ImageNet10 and COCO. For each dataset, we keep the generated repair pool, generated-image budget, source budget, checkpoint, and training schedule fixed, and remove one target field from the source-selection vocabulary and its corresponding generated-count term. `Full` uses both target fields for source selection. In each missing-field cell, the value outside parentheses is the downstream task metric after retraining.

ImageNet10: target `pink AND fabric`, overall test accuracy (%). Values in parentheses `(...)` are the selected-set Jaccard overlap with `Full` at the same ratio.

Ratio	Full	Missing color	Missing texture
0.1	73.69	72.69 (0.5604)	**73.85** (0.3904)
0.3	**74.92**	73.77 (0.6166)	73.54 (0.5842)
0.5	**75.08**	74.23 (0.6443)	74.23 (0.7330)
0.7	**75.15**	74.00 (0.6999)	74.23 (0.8826)
0.9	**75.62**	75.08 (0.8765)	74.54 (0.9668)

COCO: target `brown AND rainy`, keypoint AP. Values in parentheses `(...)` are the selected-set Jaccard overlap with `Full` at the same ratio.

Ratio	Full	Missing color	Missing weather
0.1	**0.6183**	0.6055 (0.4782)	0.6105 (0.2034)
0.3	**0.6306**	0.6221 (0.5611)	0.6214 (0.4613)
0.5	**0.6317**	0.6178 (0.7063)	0.6271 (0.6084)
0.7	0.6319	**0.6332** (0.7836)	0.6323 (0.8110)
0.9	**0.6355**	0.6237 (0.8969)	0.6244 (0.8708)

> **Vocabulary-coverage takeaway.** Missing a target field changes the selected source set most strongly at low budgets: at ratio `0.1`, Jaccard falls to `0.5604/0.3904` for missing color/texture on ImageNet10 and `0.4782/0.2034` for missing color/weather on COCO. Downstream changes are non-monotonic, however. Mean ImageNet10 accuracy changes are `−0.94/−0.81` points for missing color/texture, while mean COCO AP changes are `−0.0091/−0.0065` for missing color/weather. **Missing-field variants are lower than Full in 17 of the 20 dataset--field--ratio comparisons.** Incomplete coverage therefore changes source selection and generally lowers task performance, although the effect is not strictly monotonic at every ratio.

References

1. SafeFix: Targeted Model Repair via Controlled Image Generation. TMLR, 2026.

Add: **Official Comment** **Author AC Confidential Comment**

Official Review of Submission12417 by Reviewer KaLa

Official Review by Reviewer KaLa 15 Jun 2026, 03:38 (modified: 28 Jul 2026, 13:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Revisions (/revisions?id=12XbHqOFYG)

Summary:
This paper studies targeted vision model repair from the perspective of repair-time source-data selection. Given a diagnosed failure slice and generated repair images, the proposed AttriSelect method selects a compact subset of original training examples to accompany the generated data during retraining, aiming to reduce computational cost while preserving repair effectiveness. The method uses attribute-driven criteria, including class-conditional rarity, generated-data coverage control, and soft class balance, to better capture repair-relevant semantic slices than conventional uncertainty-, diversity-, or gradient-based active learning methods. Experiments across image classification, pose estimation, and object detection suggest the effectiveness of AttriSelect.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths.

1. The paper presents an interesting and inspiring perspective by formulating data selection as an important component of the targeted model repair process.
2. The experimental results are promising and suggest that the proposed attribute-driven selection strategy can improve repair performance across different vision tasks.

Weaknesses.

1. The problem formulation needs to be clarified. It is not always clear whether the proposed method is selecting generated repair data, original source data, or a combination of both. The descriptions in the abstract, introduction, and conclusion appear somewhat vague or potentially inconsistent on this point. The authors are encouraged to state explicitly what data pool AttriSelect operates on and how the selected data are used during retraining.
2. The categorization of prior model repair methods requires further clarification. In particular, reference [7] seems to be associated with both the first and second lines of work, which makes the taxonomy confusing. The authors should revise this discussion to make the distinctions among different categories more precise.
3. It is unclear what repair method or methods are used together with AttriSelect in the experiments. Since AttriSelect is a data selection strategy rather than a complete repair algorithm, the experimental section should clearly specify the underlying repair pipeline, retraining/fine-tuning procedure, and how the selected source samples are combined with the generated repair images.

4. The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. However, the experiments do not appear to evaluate its compatibility with multiple repair pipelines. It would strengthen the paper if the authors could test AttriSelect with different repair methods or, at least, discuss the extent to which the conclusions depend on the specific repair pipeline used in the experiments.
5. The formatting of the references should be improved. For example, the names of journals and conferences should be made consistent, either using full names or standard abbreviations throughout the reference list.

Quality: 2: not good
Clarity: 2: not good
Significance: 3: good
Originality: 3: good
Questions:

1. Could the authors clearly specify whether AttriSelect selects from the original source/training data, the generated repair data, or both at related parts?
2. Since AttriSelect is presented as a data selection component rather than a complete repair method, could the authors specify the exact repair pipeline used in each experiment?
3. The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. Could the authors provide evidence or discussion about whether AttriSelect remains effective when paired with different repair pipelines or retraining strategies?
4. Could the authors improve the categorization of prior repair methods?

Limitations:
yes

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
NA

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer KaLa (Part 1)

Edit

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

30 Jul 2026, 01:56 (modified: 30 Jul 2026, 02:37)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Revisions (/revisions?id=nlvojL88SP)

Comment:
Dear Reviewer KaLa, thank you for asking us to clarify the selected data pool, repair protocol, and compatibility with other pipelines.

This response is organized into four parts.

W1/Q1 — Selected Data Pool

> "The problem formulation needs to be clarified. It is not always clear whether the proposed method is selecting generated repair data, original source data, or a combination of both. The descriptions in the abstract, introduction, and conclusion appear somewhat vague or potentially inconsistent on this point. The authors are encouraged to state explicitly what data pool AttriSelect operates on and how the selected data are used during retraining."

Response. `AttriSelect selects only from the original source or training pool. It does not select generated repair images or rank a joint source--generated pool.` Let r denote the source-selection ratio in our tables. AttriSelect returns $\mathcal{D}^{sel}$, a subset of $\mathcal{D}^{src}$. If the source pool contains N images, the selected subset contains $\lfloor rN \rfloor$ images. In every main experiment, the upstream repair pipeline diagnoses a failure slice and constructs a generated repair set. That set contains the same 1,000 images for every selector and source ratio. The retraining set $\mathcal{D}^{repair}$ is the union of the selected source subset $\mathcal{D}^{sel}$ and the fixed generated set $\mathcal{D}^{gen}$.

AttriSelect uses attribute counts from the fixed generated set to identify values that are already well represented and avoid redundant source examples. Every ranked candidate and every selected image still comes from the source pool. We then continue training or fine-tune the diagnosis checkpoint on $\mathcal{D}^{repair}$. Therefore, changing r changes only the number and composition of real source images; the generated-image budget and membership do not change.

We will use this terminology consistently in the abstract, introduction, problem formulation, framework figure, experimental protocol, and conclusion: the upstream pipeline supplies the diagnosed slice and fixed generated set; AttriSelect selects a compact source subset; retraining uses their union. We will remove any notation that suggests a selected generated subset.

W2/Q4 — Prior-Work Taxonomy

> "The categorization of prior model repair methods requires further clarification. In particular, reference [7] seems to be associated with both the first and second lines of work, which makes the taxonomy confusing. The authors should revise this discussion to make the distinctions among different categories more precise."

Response. `The repeated citation is intentional because HiBug participates in two stages: it discovers attribute-defined failure slices and generates repair examples for them.`

We will organize the revision by pipeline stage rather than present the categories as mutually exclusive. The stages are **failure-slice discovery, repair-data construction, source-data selection, and model retraining**. HiBug can then appear in more than one stage without making the taxonomy inconsistent.

Reference [7] is HiBug, an early method that discovers human-interpretable model bugs and supports targeted repair with generated examples in computer-vision tasks. All submitted experiments place our selector inside the SafeFix repair pipeline. SafeFix addresses two limitations of direct generative augmentation: **Semantic Drift**, in which attributes unrelated to the target edit change, and **Distribution Mismatch**, in which synthetic images depart from the source-data distribution. It reduces these problems through controlled editing of real training images and VLM-based filtering.

`W2/Q4 addresses the prior-work taxonomy; W4/Q3 reports the new five-seed HiBug experiment on compatibility with another repair pipeline.`

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer KaLa (Part 2)

Edit

Official Comment by Authors 30 Jul 2026, 01:56 (modified: 30 Jul 2026, 02:36)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Revisions (/revisions?id=oTIizjQWoJ)

Comment:

W3/Q2 — Repair-Pipeline Implementation

> "It is unclear what repair method or methods are used together with AttriSelect in the experiments. Since AttriSelect is a data selection strategy rather than a complete repair algorithm, the experimental section should clearly specify the underlying repair pipeline, retraining/fine-tuning procedure, and how the selected source samples are combined with the generated repair images."

Response. `All submitted experiments place AttriSelect inside the SafeFix repair pipeline.` SafeFix follows the targeted-generation direction introduced by HiBug, using controlled edits of real source images and VLM-based filtering to construct repair images. AttriSelect leaves diagnosis and generation unchanged. It replaces only the rule for choosing the real source examples that accompany the generated repair images.

The implemented workflow is:

> **Repair-pipeline summary.**
> **Upstream SafeFix pipeline**
> Diagnosis checkpoint θ_t
> ↓
> Diagnose an attribute-defined failure slice T_{bug}
> ↓
> SafeFix controlled generation using ControlNet/Stable Diffusion 1.5, soft HED conditioning, and 30 DDIM steps
> ↓
> Fixed generated repair set $\mathcal{D}^{gen}$ (1,000 images)
> **AttriSelect source selection**
> Original source pool $\mathcal{D}^{src}$ with attribute metadata, together with attribute counts from $\mathcal{D}^{gen}$
> ↓
> AttriSelect ranks source images only
> ↓
> Selected source subset $\mathcal{D}^{sel}$
> **Repair training**
> Union of $\mathcal{D}^{sel}$ and $\mathcal{D}^{gen}$
> ↓
> Matched repair training or fine-tuning of θ_t
> ↓
> Repaired checkpoint θ_{t+1}

The generated images affect the AttriSelect score through their attribute counts, but they are never selection candidates. At source ratio r, if the source pool contains N images, the selector returns $\lfloor rN \rfloor$ images. The actual repair dataset $\mathcal{D}^{repair}$ is the union of the selected source subset $\mathcal{D}^{sel}$ and the fixed generated set $\mathcal{D}^{gen}$.

The appendix also gives the dataset-specific settings. For ImageNet10, we train the diagnosis-stage ResNet-18 for 30 epochs using cross-entropy loss, Adam, learning rate 1×10^{-4}, weight decay 1×10^{-3}, and batch size 64. Qwen3-VL supplies closed-set attribute assignments for the source images. Repair then runs for 15 epochs at learning rate 1×10^{-5}. For CelebA, COCO, and KITTI, repair uses half as many epochs as diagnosis and the same repair learning rate, 1×10^{-5}.

Within each dataset-level comparison, all selectors use the same diagnosis checkpoint, failure slice, fixed 1,000-image generated set, source ratio, task model, and repair schedule. `Only the source-subset selection rule changes.`

We will move this workflow and these settings into the experimental section so that readers do not have to assemble the repair method, data composition, and training procedure from several appendix sections.

Add: **Official Comment** **Author AC Confidential Comment**

Response to Reviewer KaLa (Part 3)

Edit

Official Comment by Authors 30 Jul 2026, 01:56 (modified: 30 Jul 2026, 02:46)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Revisions (/revisions?id=RfCHy0kEsQ)

Comment:

W4/Q3 — Compatibility with Alternative Pipelines

> "The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. However, the experiments do not appear to evaluate its compatibility with multiple repair pipelines. It would strengthen the paper if the authors could test AttriSelect with different repair methods or, at least, discuss the extent to which the conclusions depend on the specific repair pipeline used in the experiments."

> Additional point in Q3: "Could the authors provide evidence or discussion about whether AttriSelect remains effective when paired with different [...] retraining strategies?"

Alternative Repair Pipeline: HiBug

Response. HiBug [3] is an attribute-based debugging and repair framework. It uses vision--language models to find interpretable failure slices and generate targeted repair examples. We test AttriSelect inside HiBug on CelebA, replacing the SafeFix-generated repair set with a HiBug-generated set. The diagnosis checkpoint, 80,000-image source pool, source ratios, generated budget, retraining configuration, and five seeds are matched across selectors. Every row uses the same fixed HiBug-generated set; only the source-selection rule changes.

CelebA with the HiBug repair pipeline and 80,000 training images. Metric: five-seed mean FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (HiBug)
0.1	4.59	2.33	6.92	-6.20	2.87	-1.36	5.58	**7.36**
0.3	5.91	5.08	8.02	7.79	1.66	3.39	3.87	**8.07**
0.5	5.76	7.70	7.48	7.90	2.53	2.13	9.64	**12.54**
0.7	2.57	8.06	10.03	6.19	7.86	4.96	5.90	**11.16**
0.9	2.84	4.28	7.95	5.71	5.80	4.19	6.37	**13.42**

> **Key result.** `Under HiBug, AttriSelect is strongest at all five ratios, exceeding the best baseline by 0.44/0.05/2.90/1.13/5.47 FR points and by 2.00 points on average.` These comparisons should be read within the HiBug table. Changing the upstream pipeline changes the generated repair set and can shift the absolute FR of every selector.

Alternative-Generator Evidence

Response. AttriSelect requires a diagnosed semantic slice, an annotated source pool, and a fixed generated repair set. To test dependence on the main ControlNet generator, the appendix substitutes InstructPix2Pix on CelebA and ICEdit on COCO. The selectors, generated-image budget, source ratios, and repair protocol remain fixed.

CelebA with InstructPix2Pix generation. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (InstructPix2Pix)	Ours (ControlNet)
0.1	4.23	1.95	6.41	-6.65	2.34	-1.78	5.12	**7.65**	8.48
0.3	5.34	4.72	7.98	7.21	1.02	2.81	3.45	**8.12**	8.91
0.5	5.56	7.23	6.78	7.54	2.12	1.56	9.12	**11.17**	13.79
0.7	2.12	7.67	9.34	5.78	7.34	4.23	5.34	**12.13**	12.09
0.9	2.45	3.92	7.56	5.34	5.23	3.87	5.91	**13.87**	14.74

> **Key result.** `AttriSelect is the strongest source selector at every evaluated CelebA ratio under InstructPix2Pix, exceeding the strongest baseline by 2.51 FR points on average.`

COCO pose estimation with ICEdit generation. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (ICEdit)	Ours (ControlNet)
0.1	3.78	5.23	5.87	-0.34	2.34	3.12	1.56	**7.12**	6.86
0.3	3.92	8.67	8.12	9.12	6.56	4.67	6.89	**10.34**	9.86
0.5	7.91	8.45	9.67	6.91	7.82	6.54	7.23	**10.67**	10.13
0.7	6.54	10.12	9.98	9.56	9.12	6.71	8.23	**10.87**	10.18
0.9	5.12	9.87	6.98	7.82	8.12	7.54	10.12	**11.56**	11.05

> **Key result.** Under ICEdit, AttriSelect is strongest at every evaluated COCO ratio and exceeds the best baseline by **1.13 FR points on average**. `Across the two generator substitutions, AttriSelect ranks first in all 10 evaluated dataset–ratio settings.`

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Response to Reviewer KaLa (Part 4)

Official Comment by Authors 30 Jul 2026, 01:56 (modified: 30 Jul 2026, 02:45)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa
Revisions (/revisions?id=y5nbXoXLud)

Comment:

Scope of the Evidence

Scope. The empirical evidence covers one matched alternative repair pipeline (HiBug on CelebA) and two generator substitutions within the SafeFix-style workflow (InstructPix2Pix on CelebA and ICEdit on COCO).

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The appendix theory gives a conditional argument for transfer across repair pipelines and retraining strategies. The result concerns source selection and does not depend on SafeFix, HiBug, a particular image generator, or a specific optimizer. Given a diagnosed target slice, an attribute-annotated source pool, and a generated repair set, the rarity-dominates-penalty result gives conditions under which target-slice source examples remain ranked above non-target examples after accounting for the generated-data distribution. Let B_c denote the selection budget allocated to class c, and let T_c denote the set of target-slice source examples in that class. The deterministic enrichment result then guarantees at least $\min B_c, |T_c|$ selected target-slice examples within class c, whereas an attribute-agnostic selector requires its uncertainty or coverage signal to be separately aligned with that slice.

These guarantees apply before model updating, so the selected subset can feed checkpoint fine-tuning, continued training, or retraining from scratch. Under the appendix's repair-utility condition, the enrichment result gives expected repair utility at least as high as random or stratified selection under the same class allocation, with a strict advantage when the enrichment gap is strict. `Changing the repair pipeline or update rule affects the data counts and how the additional target support translates into the final metric; it does not remove the selection mechanism.` This is a conditional selection-level argument, not a guarantee for every optimizer, checkpoint, or data-weighting scheme.

W5 — Reference Formatting

> “The formatting of the references should be improved. For example, the names of journals and conferences should be made consistent, either using full names or standard abbreviations throughout the reference list.”

Response. We standardized conference and journal names throughout the reference list using NeurIPS, ICLR, ICML, CVPR, ICCV, ECCV, WACV, AAAI, FAccT, and JMLR. We also corrected title and venue capitalization, protected proper names such as ImageNet, COCO, KITTI, and SafeFix, and fixed the malformed author field in the PPAL reference. The bibliography now uses one format.

Additional Evidence from Other Reviewer Responses

Several experiments in the other responses bear directly on KaLa's questions. The hG6j response gives every method the same ImageNet10 attributes and target-slice definition. AttriSelect leads at all five source ratios by 0.59 accuracy points on average over the best control.

The yyPZ response varies repair training from 15 to 60 epochs. AttriSelect remains strongest in all eight configurations. The Area Chair response reports source-only ImageNet10 experiments that train a new ResNet-18 rather than fine-tune a repair checkpoint. AttriSelect leads at all five one-shot budgets, and AttriSelect-AL improves the six-round Curve Mean by 1.40 points over BADGE.

The other responses report five-seed studies of slice granularity, alternative attribute sources, and incomplete vocabularies. `The observed advantage is therefore not confined to one information budget, repair schedule, checkpoint–update mode, slice definition, or attribute source.`

References

1. SafeFix: Targeted Model Repair via Controlled Image Generation. TMLR, 2026.
2. Actively Selecting Annotations Among Objects and Attributes. ICCV, 2011.
3. HiBug: On Human-Interpretable Model Debug. NeurIPS, 2023.

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